

## 國立中山大學 114學年度第3學期 課程教學大綱

## National Sun Yat-sen University 114Academic year Course syllabus

|                              |                 |                          |    |                     |        |
|------------------------------|-----------------|--------------------------|----|---------------------|--------|
| 中文名稱<br>Course name(Chinese) | 電子學（一）          |                          |    | 課號<br>Course Code   | EE2100 |
| 英文名稱<br>Course name(English) | ELECTRONICS (I) |                          |    |                     |        |
| 課程類別<br>Type of the course   | 講授類             | 必選修<br>Required/Selected | 選修 | 系所<br>Dept./faculty | 電機工程學系 |
| 授課教師<br>Instructor           | 馬誠佑             |                          |    | 學分<br>Credit        | 3      |

## 課程大綱 Course syllabus

1. Electronics and Semiconductors (chapter 1)
2. Diodes (chapter 3)
3. Bipolar Junction Transistors (BJTs) and Its Amplifier Circuit (chapter 4+6)
4. Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) and Its Amplifier Circuit (chapter 4+6)

## 課程目標 Objectives

本課程介紹基本電子元件的運作原理及應用，使學生習得基礎半導體物理、二極體、雙極性接面電晶體及金氧半場效電晶體的知識，對電子相關進階課程建立觀念。

## 授課方式 Teaching methods

## 課堂講授

評分方式（評分標準及比例）Evaluation (Criteria and ratio) [等第制單科成績對照表 letter grading reference](#)

- 1.第一次期中考(7/29)：30%
- 2.第二次期中考(8/11)：35%
- 3.期末考(8/18)：35%

## 參考書/教科書/閱讀文獻 Reference book/ textbook/ documents

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| 序號  | 作者                          | 書名                                     | 出版社                     | 出版年             | 出版地             | ISBN#         |
|-----|-----------------------------|----------------------------------------|-------------------------|-----------------|-----------------|---------------|
| No. | Author                      | Title                                  | Publisher               | Year of publish | Publisher place | ISBN#         |
| 1   | A. S. Sedra and K. C. Smith | Microelectronic Circuits (8th edition) | Oxford University Press | 2021            | USA             | 9780190853501 |

## 課程時數規劃 Course Hour Planning

本校自114學年度起實施學期16週，課程（含期末考試）應於16週內完成。學分之計算仍以1學分18小時為原則。教師課程時數安排得選擇「16週+自主學習規劃」或「16週+實體上課規劃」。

Starting from the 114th academic year, the university will implement a 16-week course schedule, and all courses (including final examinations) must be completed within this 16-week time frame while maintaining the standard of 18 hours of instruction per credit. Instructors can choose between “16-weeks + Alternative learning periods” or “16-weeks + In-person classes.”

☒ 本門課程為「16週+自主學習規劃」：教師需於「A.每週課程內容及預計進度」欄位填寫16週課程進度，並於「B.自主學習規劃」欄位填寫每1學分2小時學生自主學習內容。

16 weeks + alternative learning periods: The instructor will include a 16-week course plan in the weekly scheduled progress section (16 hours of instruction per credit) and provide details of the learning plan (two hours of activity per credit) in the alternative learning period section.

☐ 本門課程為「16週+實體上課規劃」：教師需於「A.每週課程內容及預計進度」欄位填寫16週課程進度，並於「C.實體上課規劃」填寫2次授課內容及主題，且於學期16週內完成。

16 weeks + in-person classes: The instructor will include a 16-week course plan in the weekly scheduled progress section (16 hours of instruction per credit) and specify the content and topics of the 2 in-person classes in the in-person class plan section, and must be completed within this 16-week time frame.

A.每週課程內容及預計進度 Weekly scheduled progress

|                                                                                                               |                       |                                |
|---------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------|
| 全英課程之授課內容及主題應以英文或雙語呈現                                                                                         |                       |                                |
| For courses taught entirely in English, the content and topics should be presented in English or bilingually. |                       |                                |
| 週次                                                                                                            | 日期                    | 授課內容及主題                        |
| Week                                                                                                          | Date                  | Content and topic              |
| 1                                                                                                             | 2026/07/12~2026/07/18 | Electronics and Semiconductors |
| 2                                                                                                             | 2026/07/19~2026/07/25 | Diodes & BJTs                  |
| 3                                                                                                             | 2026/07/26~2026/08/01 | 1st Midterm (7/29) & BJTs      |
| 4                                                                                                             | 2026/08/02~2026/08/08 | BJTs & MOSFETs                 |
| 5                                                                                                             | 2026/08/09~2026/08/15 | 2nd Midterm (8/11) & MOSFETs   |
| 6                                                                                                             | 2026/08/16~2026/08/22 | Final exam (8/18) & 彈性學習       |

B.自主學習規劃 Alternative learning periods

課程規劃學生自主學習內容（每1學分2小時）  
Alternative learning periods planned for the course (with each credit corresponding to two hours of activity)

本門課程規劃學生彈性或自主學習內容（每1學分2小時）：  
Alternative learning periods planned for the course (each credit corresponds to two hours of activity):

| 學生自主學習活動<br>Alternative learning periods                                                                                        | 勾選或填寫規劃內容<br>Place a check in the appropriate box or provide details | 時數<br>Number of hours |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------|
| 學生分組實作及討論<br>Group work and discussion                                                                                          | <input type="checkbox"/>                                             |                       |
| 參與課程相關作業、作品、實驗<br>Participation in course-related assignments, work, or experiments                                             | <input type="checkbox"/>                                             |                       |
| 參與校內外活動（研習營、工作坊、參訪）或競賽<br>Participation in on- or off-campus activities (e.g., seminars, workshops, and visits) or competitions | <input type="checkbox"/>                                             |                       |
| 課外閱讀<br>Extracurricular reading                                                                                                 | <input type="checkbox"/>                                             |                       |
| 線上數位教材學習<br>Learning with online digital learning materials                                                                     | <input checked="" type="checkbox"/><br>至Youtube進行線上數位教材複習            | 6                     |
| 其他（請填寫規劃內容）<br>Other (please provide details)                                                                                   | <input type="checkbox"/>                                             |                       |

C.實體上課規劃 In-Person Class Plan

若無規劃學生自主學習，則請教師規劃2次實體上課(每1學分2小時)，應於學期16週內完成實體上課（含期末考試），實體上課時間由師生自行討論，得利用週三下午4-7點或其他時段進行。  
If there are no alternative learning periods planned for the course, the instructor should plan 2 in-person classes (2 hours of activity per credit).in-person class must be completed within this 16-week time frame(including final examinations). In-person Class schedules can be arranged through discussions between instructors and students, utilizing Wednesday 4:00PM-7:00PM or other suitable time slots.

\* 第一次實體上課內容及主題 (Content and topic for the first In-Person class)：  
\* 第二次實體上課內容及主題 (Content and topic for the second In-Person class)：

課業討論時間 Office hours

時段1 Time period 1:  
時間 Time：星期二11:00~13:00  
地點 Office/Laboratory：工EC7023-2  
時段2 Time period 2：  
時間 Time：星期三11:00~13:00  
地點 Office/Laboratory：工EC7023-2

系所學生專業能力/全校學生基本素養與核心能力 basic disciplines and core capabilities of the dcpartment and the university

| 系所學生專業能力/全校學生基本素養與核心能力<br>basic disciplines and core capabilities of the department | 課堂活動與評量方式<br>Class activities and evaluation |        |                                 |                                     |                                |                                    |                                    |                            |                   |                                                        | 課外閱讀<br>Outside reading. |
|-------------------------------------------------------------------------------------|----------------------------------------------|--------|---------------------------------|-------------------------------------|--------------------------------|------------------------------------|------------------------------------|----------------------------|-------------------|--------------------------------------------------------|--------------------------|
|                                                                                     | 本課程欲培養之能力與素養<br>This course                  | 紙筆考試或測 | 課堂討論(含個案討論)<br>Group discussion | 個人書面報告、作業、作品、實驗<br>Individual paper | 群組書面報告、作業、作品、實驗<br>Group paper | 個人口頭報告<br>Indivisual presentation. | 群組口頭報告<br>Group oral presentation. | 課程規劃之校外參訪及實習<br>Off-campus | 證照/檢定<br>License. | 參與課程規劃之校內外活動及競賽<br>Participate in off-campus/on-campus |                          |

| and the university                                                                                                                                       | enables students to achieve. | 驗 Test. 紙筆考試或測驗 Test. 紙筆考試或測驗 Test. | (case analysis). | report/ assignment/ work or experiment. | report/ assignment/ work or experiment. |  |  | visit and intership. |  | activities and competitions. |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------|------------------|-----------------------------------------|-----------------------------------------|--|--|----------------------|--|------------------------------|---|
| ※系所學生專業能力 Basic disciplines and core capabilities of the department                                                                                      |                              |                                     |                  |                                         |                                         |  |  |                      |  |                              |   |
| 1.能夠整合、組織電機專業理論來分析、表達問題之能力。 1. Be able to analyze, express and solve problems by integrating Electrical Engineering knowledge.                           | V                            | V                                   | V                |                                         |                                         |  |  |                      |  |                              | V |
| 2.能夠運用電機專業知識解決及實作電機工程問題之能力。 2. Be able to apply professional knowledge of electrical engineering to solve and implement electrical engineering problems. | V                            | V                                   | V                |                                         |                                         |  |  |                      |  |                              | V |
| 3.具備分工、協調、重視團隊合作精神、遵守工程倫理以達成工作目標之能力。 3. Be able to achieve the goal by working with team and following engineering ethics.                               |                              |                                     |                  |                                         |                                         |  |  |                      |  |                              |   |
| 4.能夠激發自己潛能、融合他人智慧，具備獨立思考以及研究創新之能力。 4. Be able to find out his/her own potential, think independently and innovatively.                                   | V                            | V                                   | V                |                                         |                                         |  |  |                      |  |                              | V |
| 5.具備吸收電機新知、掌握國際發展趨                                                                                                                                       |                              |                                     |                  |                                         |                                         |  |  |                      |  |                              |   |

|                                                                                                                                                   |   |   |   |  |  |  |  |  |  |  |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|--|--|--|--|--|--|--|---|
| 勢，隨時接受競爭挑戰之能力。5. Be able to absorb new knowledge about motors, grasp international development trends, and readily accept competitive challenges. |   |   |   |  |  |  |  |  |  |  |   |
| ※全校學生基本素養與核心能力 Basic disciplines and core capabilities of the university                                                                          |   |   |   |  |  |  |  |  |  |  |   |
| 1.表達與溝通能力。1. Articulation and communication skills                                                                                                |   |   |   |  |  |  |  |  |  |  |   |
| 2.探究與批判思考能力。2. Inquisitive and critical thinking abilities                                                                                        | V | V | V |  |  |  |  |  |  |  | V |
| 3.終身學習能力。3. Lifelong learning                                                                                                                     | V | V | V |  |  |  |  |  |  |  | V |
| 4.倫理與社會責任。4. Ethics and social responsibility                                                                                                     |   |   |   |  |  |  |  |  |  |  |   |
| 5.美感品味。5. Aesthetic appreciation                                                                                                                  |   |   |   |  |  |  |  |  |  |  |   |
| 6.創造力。6. Creativity                                                                                                                               |   |   |   |  |  |  |  |  |  |  |   |
| 7.全球視野。7. Global perspective                                                                                                                      |   |   |   |  |  |  |  |  |  |  |   |
| 8.合作與領導能力。8. Team work and leadership                                                                                                             |   |   |   |  |  |  |  |  |  |  |   |
| 9.山海胸襟與自然情懷。9. Broad-mindedness and the embrace of nature                                                                                         |   |   |   |  |  |  |  |  |  |  |   |

本課程與SDGs相關項目：The course relates to SDGs items:

- ☐ SDG1-消除貧窮(No Poverty)
- ☐ SDG2-消除飢餓 (Zero Hunger)
- ☐ SDG3-良好健康與福祉(Good Health and Well-being)
- ☐ SDG4-教育品質(Quality Education)
- ☐ SDG5-性別平等(Gender Equality)
- ☐ SDG6-乾淨水源與公共衛生(Clean Water and Sanitation)
- ☐ SDG7-可負擔乾淨能源(Affordable and Clean Energy)

- |                                     |                                                          |
|-------------------------------------|----------------------------------------------------------|
| <input type="checkbox"/>            | SDG8-優質工作與經濟成長(Decent Work and Economic Growth)          |
| <input checked="" type="checkbox"/> | SDG9-工業、創新和基礎建設(Industry,Innovation and Infrastructure)  |
| <input type="checkbox"/>            | SDG10-減少不平等(Reduced Inequalities)                        |
| <input type="checkbox"/>            | SDG11-永續城市(Sustainable Cities and Communities)           |
| <input type="checkbox"/>            | SDG12-責任消費與生產(Responsible Consumption and Production)    |
| <input type="checkbox"/>            | SDG13-氣候行動(Climate Action)                               |
| <input type="checkbox"/>            | SDG14-海洋生態(Life Below Water)                             |
| <input type="checkbox"/>            | SDG15-陸域生態(Life on Land)                                 |
| <input type="checkbox"/>            | SDG16-和平、正義和穩健的制度(Peace,Justice And Strong Institutions) |
| <input type="checkbox"/>            | SDG17-促進目標實現的全球夥伴關係(Partnership for the Goals)           |
| <input type="checkbox"/>            | 本課程和SDGS無關                                               |

本課程校外實習資訊: This course is relevant to internship:

☐ 本課程包含校外實習（本選項僅供統計使用，無校外實習者，得免勾記）  
The course includes internship.(For statistical use only. If the course without internship, please ignore this item.)

實習定義：規劃具有學分或時數之必修或選修課程，且安排學生進行實務與理論課程實習，於實習終了取得考核證明繳回學校後，始得獲得學分；或滿足畢業條件者。（一般校內實習請勿勾選此欄位）

Internship: The required or elective courses should include credits and learning hours. Students should participate in the corporative company or institution to practice and learn the real skills. An internship certification must be handed in at the end of internship to get the credits or to fulfil the graduation requirements.

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